A Project Presentation on

ABHA-Based Inter-Hospital Medical Record Exchange

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Presentation Outline

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1.Motivation

The project aims to simplify secure medical record sharing between hospitals using ABHA-linked health information. It focuses on improving healthcare coordination and making patient history easier to access with consent.

- To improve accessibility of medical records.
- To reduce delays in accessing patient history.
- To avoid unnecessary repetition of medical tests.
- To support faster review through AI-assisted summarization.

2.Problem Statement

Medical records are often stored in different hospitals, making it difficult for doctors to access a patient's complete medical history. A secure, consent-based system is required to exchange ABHA-linked medical records efficiently between hospitals.

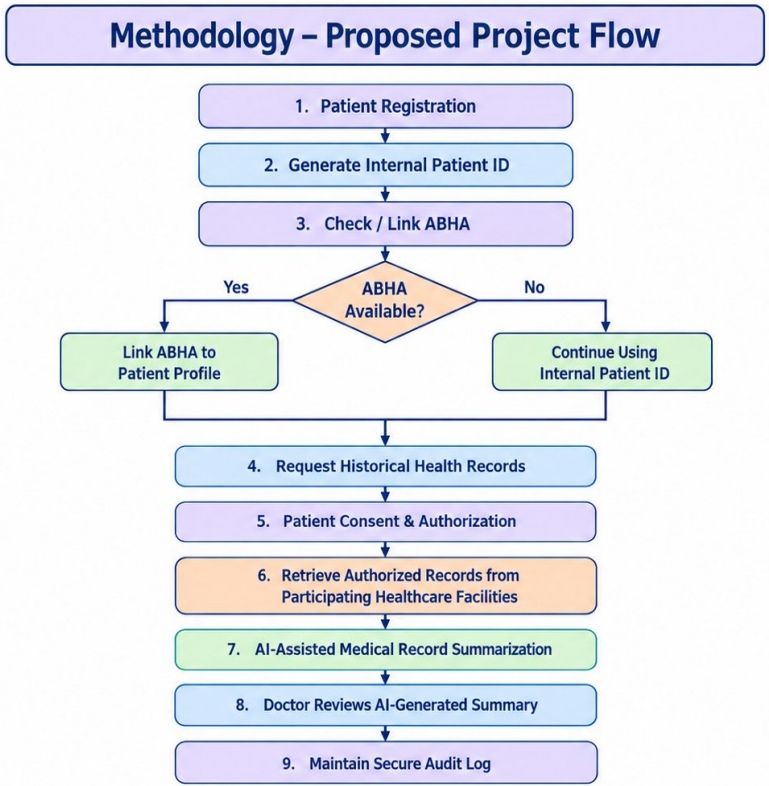
3. Existing System

Existing System	Literature Review (Related Research Papers)	Limitations	Gaps Identified
Hospitals store medical records separately.	1. **The Ayushman Bharat Digital Mission (ABDM): Making of India's Digital Health Story (2023)** Sharma, R., Rohatgi, J. & Singh, A. *Journal of Health Management*, 11(1), 10-23. DOI: 10.1007/s40012-023-00375-0 https://doi.org/10.1007/s40012-023-00375-0	• Records remain fragmented across different systems.	• Need for ABHA-linked record exchange between hospitals.
Patients carry reports between hospitals.	2. **Inception of the Indian Digital Health Mission: Connecting the Dots (2023)** Raj, A., Dananjayan, P. & Agarwal, S. *Health Care Science*, 2(5), 100-110. DOI: 10.1002/hcs2.67 https://doi.org/10.1002/hcs2.67	• Sharing needs proper authorization and verification.	• Secure, consent-based sharing of medical records.
Doctors review multiple medical reports.	3. **India's Evolving Digital Health Strategy (2024)** Narayan, B., Bhushan, S. & Schulman, E. *npj Digital Medicine*, 7, Article 284. DOI: 10.1038/s41746-024-01279-2 https://doi.org/10.1038/s41746-024-01279-2	• Reviewing long medical histories takes time.	• AI-assisted summaries for quick and efficient access to patient history.

4. Objectives

1. To enable secure exchange of medical records between hospitals.
2. To use ABHA for patient identification.
3. To ensure patient consent before sharing records.
4. To reduce delays in accessing medical history.
5. To generate AI-assisted medical summaries.
6. To improve healthcare coordination.

5. Methodology



6. Software Hardware Requirements

Software Requirements

- **Operating System** Windows 10/11 or Ubuntu Linux
- **Frontend** HTML, CSS & JavaScript (Web Portal)
- **Backend** Python with Flask
- **Database** MySQL

AI & Machine Learning :- Python

NLP (Natural Language Processing)
Scikit-learn *(if ML features are implemented)*
Pandas & NumPy

- **APIs & AI Integration**

OpenAI API / Gemini API *(for AI-assisted medical record summarization)*
REST API
ABDM-compatible APIs *(for ABHA integration, subject to access and approval)*

- **Development Tools**

Visual Studio Code
Postman
Git & GitHub

6. Software Hardware Requirements

Hardware Requirements:

• **Development Machine**
- Intel Core i3 (10th Gen or above) / AMD Ryzen 3 or above
- Minimum 8 GB RAM
Recommended 16 GB RAM
256 GB SSD (Minimum)
- Internet Connection for API Integration

• **Server (Deployment)**
- 2 vCPUs or higher
- 4 GB RAM or higher
- 50 GB SSD Storage
- Linux-based Cloud Server

• **Client Devices**
- Desktop Computer
- Laptop
- Android Smartphone
- Tablet (Optional)

• **Network**
- Stable Broadband / Wi-Fi
- HTTPS-enabled Internet Connection

7. UML DAIGRAM

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8. Conclusion

The proposed system integrates ABHA-based health record exchange with AI-assisted medical record summarization. It aims to improve healthcare data accessibility, security, and efficiency while supporting informed clinical decisions through consent-based information sharing.

9. References

1. **Ayushman Bharat Digital Mission (ABDM)**
   https://abdm.gov.in/

2. **Python Official Documentation**
   https://docs.python.org/3/

3. **Flask Official Documentation**
   https://flask.palletsprojects.com/en/stable/

4. **OpenAI API Documentation**
   https://developers.openai.com/api/reference/python

Question and Answer